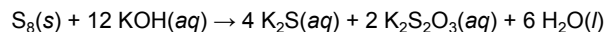


Elemental sulfur reacts with hot aqueous potassium hydroxide as shown in the reaction below.



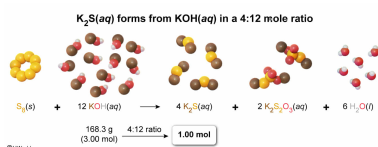
If 168.3 g of potassium hydroxide is consumed in a particular reaction, how many moles of potassium sulfide are formed in the reaction? (Note: The molar mass of potassium hydroxide is 56.1 g/mol.)

- ☐ A. 0.50 mol [20%]
☒ B. 1.00 mol [66%]
☐ C. 9.00 mol [8%]
☐ D. 12.0 mol [4%]

Correct

66% answered correctly

Explanation:



The **molar mass** of a compound is the amount of **mass contained in 1 mole** of the compound (ie, the grams per mole), and can be used to convert the mass of a compound to the equivalent number of moles. As such, if 168.3 g of potassium hydroxide (KOH) is consumed in the given reaction, the moles of KOH in the reacting mass can be found by dividing by the molar mass of KOH:

$$168.3 \text{ g KOH}(\text{aq}) \times \frac{1 \text{ mol KOH}(\text{aq})}{56.1 \text{ g KOH}(\text{aq})} = 3.00 \text{ mol KOH}(\text{aq})$$

The number of moles of one chemical species is related to the number of moles of another chemical species in a reaction by the **mole ratios** of the **stoichiometric coefficients** from a **balanced reaction equation**. The given balanced reaction equation shows that 4 mol of potassium sulfide (K_2S) is formed for every 12 mol of KOH that is consumed. Applying this mole ratio, the number of moles of $\text{K}_2\text{S}(\text{aq})$ formed from the conversion of 3.00 mol of $\text{KOH}(\text{aq})$ in the reaction is:

$$3.00 \text{ mol KOH}(\text{aq}) \times \frac{4 \text{ mol K}_2\text{S}(\text{aq})}{12 \text{ mol KOH}(\text{aq})} = 3.00 \text{ mol KOH}(\text{aq}) \times \frac{4 \text{ mol K}_2\text{S}(\text{aq})}{3 \cdot 4 \text{ mol KOH}(\text{aq})} = 1.00 \text{ mol K}_2\text{S}(\text{aq})$$

(Choice A) A total of 0.50 mol of potassium thiosulfate ($\text{K}_2\text{S}_2\text{O}_3$) is formed in the reaction, but the question asks for the moles of *potassium sulfide* (K_2S).

(Choice C) A total of 9.00 mol is calculated if the mole ratio $\frac{4 \text{ mol K}_2\text{S}(\text{aq})}{12 \text{ mol KOH}(\text{aq})}$ is mistakenly inverted as $\frac{12 \text{ mol KOH}(\text{aq})}{4 \text{ mol K}_2\text{S}(\text{aq})}$ during the last step of the calculation.

(Choice D) A total of 12.0 mol is calculated by mistakenly multiplying the 3.00 moles of $\text{KOH}(\text{aq})$ by 4 (ie, the stoichiometric coefficient of K_2S) but not accounting for the stoichiometric coefficient of $\text{KOH}(\text{aq})$. The reaction occurs in a 4:12 $\text{K}_2\text{S}(\text{aq})$: $\text{KOH}(\text{aq})$ mole ratio, not a 4:1 $\text{K}_2\text{S}(\text{aq})$: $\text{KOH}(\text{aq})$ mole ratio.

Educational objective:

The molar mass (the amount of mass in 1 mole of a compound) permits conversions between a given amount of mass and the number of moles it contains. Stoichiometric mole ratios from a balanced reaction equation relate the number of moles of one chemical species to the moles of another chemical species in a reaction.

Time spent: 0 sec
 Subject:
 General Chemistry

QID: 404125
 System:
 Interactions of Chemical Substances